

Database Design Document (DDD)

ScrapFlow AI – Version 1.0

1. Document Overview

Purpose

This document defines the complete database structure for ScrapFlow AI.

It specifies:

- Tables
- Relationships
- Constraints
- Indexes
- Data flow
- Business rules

This document serves as the foundation for:

- MySQL Database
 - FastAPI Models
 - API Development
 - Business Logic
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2. Database Overview

Database Name

scrapflow_db

Database Type

MySQL 8+

Design Goals

- Scalability

- Data Integrity
 - Performance
 - Easy Reporting
 - SaaS Multi-user Support
-

3. Entity Relationship Overview

Users

|

└─ Batches

|

└─ Materials

|

└─ Expenses

Materials

|

└─ Inventory

Buyers

|

└─ Sales

Sales

|

└─ Materials

4. Users Table

Purpose

Store user accounts.

Table Name

users

Fields

Field	Type	Constraints
id	BIGINT	PK, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(255)	UNIQUE
password_hash	VARCHAR(255)	NOT NULL
role	ENUM	owner, worker
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

5. Batches Table

Purpose

Store daily scrap collection batches.

Table Name

batches

Fields

Field	Type
id	BIGINT PK
user_id	BIGINT FK
batch_name	VARCHAR(100)

Field	Type
collection_date	DATE
collection_location	VARCHAR(255)
collection_cost	DECIMAL(10,2)
fuel_cost	DECIMAL(10,2)
status	ENUM
created_at	TIMESTAMP

Status Values

Collected

Sorting

Sorted

Partially Sold

Sold

6. Materials Table

Purpose

Store segregated materials.

Table Name

materials

Fields

Field	Type
id	BIGINT PK
batch_id	BIGINT FK

Field	Type
material_type	VARCHAR(50)
weight	DECIMAL(10,2)
created_at	TIMESTAMP

Material Types

Copper

Aluminum

Steel

Iron

Brass

Plastic

Other

7. Inventory Table

Purpose

Maintain current stock.

Table Name

inventory

Fields

Field	Type
id	BIGINT PK
material_type	VARCHAR(50)
available_weight	DECIMAL(10,2)

Field	Type
updated_at	TIMESTAMP

8. Buyers Table

Purpose

Store buyer information.

Table Name

buyers

Fields

Field	Type
id	BIGINT PK
name	VARCHAR(150)
phone	VARCHAR(20)
address	TEXT
notes	TEXT
created_at	TIMESTAMP

9. Sales Table

Purpose

Store material sales.

Table Name

sales

Fields

Field	Type
id	BIGINT PK
buyer_id	BIGINT FK
material_id	BIGINT FK
quantity	DECIMAL(10,2)
price_per_kg	DECIMAL(10,2)
total_amount	DECIMAL(10,2)
sale_date	DATE
created_at	TIMESTAMP

10. Expenses Table

Purpose

Store operational expenses.

Table Name

expenses

Fields

Field	Type
id	BIGINT PK
batch_id	BIGINT FK
expense_type	VARCHAR(50)
amount	DECIMAL(10,2)
notes	TEXT
created_at	TIMESTAMP

Expense Types

Fuel

Transport

Maintenance

Miscellaneous

11. Reports View Tables

These are generated dynamically.

No physical table required.

Daily Profit Report

Revenue - Expenses

Monthly Profit Report

Total Revenue - Total Expenses

Inventory Report

Current Inventory Snapshot

12. Relationships

User → Batch

One User



Many Batches

Batch → Materials

One Batch



Many Materials

Batch → Expenses

One Batch



Many Expenses

Buyer → Sales

One Buyer



Many Sales

Material → Sales

One Material



Many Sales

13. Foreign Keys**batches**

user_id

REFERENCES users(id)

materials

batch_id

REFERENCES batches(id)

expenses

batch_id

REFERENCES batches(id)

sales

buyer_id

REFERENCES buyers(id)

material_id

REFERENCES materials(id)

14. Business Constraints

Constraint 1

Inventory cannot be negative.

Constraint 2

Sale quantity cannot exceed available inventory.

Constraint 3

Batch status must be valid.

Constraint 4

Material weight must be greater than zero.

Constraint 5

Expense amount must be greater than zero.

Constraint 6

Sales amount must be greater than zero.

15. Indexing Strategy

Users

email

Batches

collection_date

batch_name

Materials

material_type

Buyers

phone

name

Sales

sale_date

16. Inventory Update Pipeline

Create Batch



Add Materials



Materials Stored



Inventory Updated

17. Sales Pipeline

Select Buyer



Select Material



Check Inventory



Create Sale



Reduce Inventory



Store Sale

18. Profit Calculation Pipeline

Revenue



Fetch Expenses



Calculate Profit



Store Dashboard Data

19. Dashboard Data Sources

Dashboard Card	Source
Today's Sales	sales
Today's Expenses	expenses
Today's Profit	sales + expenses
Inventory Value	inventory
Active Batches	batches
Total Inventory Weight	inventory

20. Future Tables (Version 2)

voice_notes

id
user_id
audio_path
transcription
created_at

ai_queries

id
user_id
query
response
created_at

notifications

id

user_id

message

status

created_at
